

[L2] Chapters 3 & 4 - Creating the Computer

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- There were computers being built outside the US and UK as well:
 - Konrad Zuse built a computer in his parents' home in 1938 which "seemed to be working"
 - In Japan the Yamashita statistical machine was built 1944-1948 because the Japanese government relied on IBM machines before the war. However, with the start of the war, they couldn't get these anymore, so they had to build their own.

Cold War Science

- After WW2, the need for scientific calculations exploded, with big players such as NASA or atomic power.
- In continental Europe, this happened as well
 - Even though countries were devastated after the war, countries were rebuilding
 - Building a computer was not that expensive relatively
 - There was a mix of fatalism and optimism. Everything had been torn to pieces and it seemed like it had come to no good. On the other hand, there were people who were optimistic that they could work towards a better society and achieve something great.
- 1946 the Mathematisch centrum was established in Amsterdam
 - There were several ties to the Moore School
- Cold war Europe was not following in every respect:
 - Europe wanted to compete with the flow of American products and technology flowing into Europe
 - A need was felt to have their own products, technologies, ideas, associations and, of course, computers
 - Within every European country there were scientists trying to build machines. They did not want to be dependent on America
- Certain components that were mass-produced for the war were now available en-masse because of the end of the war

Dutch Dinosaurs

Why?

- Aeronautics
- Land reclamation projects: H.A. Lorentz and C. Lely needed large calculations for the Zuiderzee land reclamations. They designed the calculations and had human computers perform them
- Process control: Johan van Veen designed analog machines with the purpose of solving hydrology problems
- Deltar was a machine used to perform calculations for the Zeeland land reclamations

Amsterdam:

- Aad van Wijngaarden went to England to see what progress was being done there and came back to Amsterdam with ideas
 - He assembled parts and people and started building a computer
 - June 21, 1952 he presented the ARRA to the minister of sciences, with a program that could generate a random number (quite a feat in computing)
 - The machine worked during the presentation, but never afterwards.
- The ARRA II was assembled from transistors instead of relays and vacuum tubes. It was quite successful and worked for a while
- The ARMAC came from the ARRA in 1956
- The Electrologica X1 was built on the ARRA II design in 1956 and was a spin-off of the mathematical centrum

Delft

- Willem van der Poel was from the school of fiber optics and was interested in having a machine that could calculate complex constellations of optical lenses
- He constructed the ARCO (Automatic Relais Calculator for Optical problems). It was nicknamed Testudo (turtle) because it was so incredibly slow. This was in 1950
- Delft had connections with the Dutch telephone company PTT and they created the ZERO for them in 1951. The PTT wanted a machine that could supplant the operator. They used it under the name of PTERA
- 1955 they designed the ZEBRA (Zeer Eenvoudig Binair Reken Apparaat) and it was produced by STANTEC in the UK, because, in the Netherlands, Philips was not interested.

Eindhoven

- Philips was building machines
- 1953-1956 the PETER was built. Wim Nijenhuis was the chief engineer and wanted a machine that could test electronics and perform calculations with the results. They did not mass produce these machines, because they had an agreement with IBM, whereby IBM would produce computers and buy Philips parts for them.
- Philips built PASCAL (Philips Akelig Snelle Calculator) and STEVIN (Snel Tel- En Vermenigvuldigings Instrument) that were used within

Observations:

- These machines were built in order to function.
- They were useful for very specific jobs or for applications where there was absolutely no room for human errors
- The names and the machines proved the European necessity to establish something in electronic computing. Machines became an element of national pride.

Automation in Culture / Media

- In the book "1984", George Orwell described a world that would go berserk on systems thinking (mathematics, science, etc). Technology could cause a shift in power structures. It portrayed totalitarianism through surveillance, technology.
- All major projects had to be sold. Documentaries were made about atomic power, for example. ("A is for Atom" made by General Electric). A Disney documentary about space travel featured Werner von Braun with one of his Space station models
- In "The Adventures of Tintin", when he asks a professor how he knows that his calculations are correct for the trajectory of a space rocket, a stack of papers is thrown at him. Calculations were done on paper

After 1955, computers start emerging in popular culture

- In "Desk set" (1957), an office is "invaded" by machines that threatened to make the human workers obsolete and out of a job

The Golden Age of Science Fiction

- One of the most important authors was Isaac Asimov (1920-1992)
 - I, Robot (1950)
 - Foundation (1951)
 - The last question (1956)

Changes in Movie Culture:

- Change in storage medium
 - Celluloid to DVD to Cloud
 - This changed the way in which people watch movies. Films became more widely available
 - This was disastrous for movie theatres, because people could watch at home
 - Cloud storage also shut down DVD rental stores
- Change in theme:
 - "2001" (1968) had a theme of space travel. The AI went berserk => fault in the software, in the machine
 - In the 1980's movies ("Blade Runner", "Wargames"), it's not the software's fault. The human factor is in question (unthinkable in the late 60's)
 - In 2013, the movie "Her" is about a man who falls in love with his machine. (unthinkable in the 80's)
- Special Effects:
 - From stop motion and perspective tricks to special effects and computer animation
- Change in stunts:
 - From stunts being performed by stuntmen and stuntwomen to them being done by the actors in front of greenscreens